



Executive Report: Integrated Automated Incident Response Framework (EDR & SOAR)

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Introduction

As global cyber threats escalate and organizations shift heavily toward digital platforms, adopting advanced security solutions that combine intelligence, automation, and real-time analysis has become essential. This project addresses a critical gap in incident detection and response by integrating two leading platforms—LimaCharlie and Tines—delivering a practical step toward a more secure, intelligent digital ecosystem.

The project establishes a hands-on framework for integrating Endpoint Detection and Response (EDR) systems with Security Orchestration, Automation, and Response (SOAR) platforms. The objective is to build a fully automated model capable of:

- Gathering and analyzing threat data across multiple sources
- Executing automated response actions without requiring continuous manual intervention
- Empowering security teams to make strategic, data-driven decisions
- Strengthening cross-departmental collaboration within the organization

Playbook workflows were mapped out using tools like Draw.io and Slack to illustrate various incident response scenarios—including automated endpoint isolation, real-time alerting, and centralized incident logging. To guarantee operational readiness and long-term viability, the system's effectiveness was thoroughly validated in an environment simulating real-world enterprise conditions.



Project Overview

The core of this project lies in developing and implementing an integrated model combining LimaCharlie and Tines. The primary objective is to establish a practical, automated cybersecurity architecture that enables real-time threat detection and intelligent incident response, minimizing the need for constant manual intervention.

The approach hinges on unifying Endpoint Detection and Response (EDR) solutions with Security Orchestration, Automation, and Response (SOAR) platforms. Together, these technologies foster an environment driven by automation, instant threat interaction, and data-backed analysis.

Technical Objectives

- **Rapid Threat Detection:** Leverage LimaCharlie's ability to gather and analyze endpoint telemetry to detect threats swiftly and generate real-time alerts upon identifying suspicious activity.
- **Automated Incident Response:** Design automated playbooks in Tines to execute remediation actions seamlessly—such as isolating compromised endpoints, dispatching alerts, and centralizing event logging.
- **Streamlined Team Collaboration:** Connect both platforms to communication tools like Slack to ensure fast information sharing and efficient task coordination across security teams.
- **Multi-Source Data Analysis:** Aggregate security data from diverse sources to enhance visibility into malicious behavior and inform strategic decision-making.

Practical Application & Operational Focus

Designed for real-world usability, the project offers a viable deployment model for organizations operating without heavily funded infrastructures by prioritizing:

- **Intuitive Workflow Design:** Creating accessible interfaces that allow operational teams to handle incidents smoothly.
- **Non-Functional Requirements:** Ensuring system reliability, scalability, and straightforward administration.
- **Clear Process Documentation:** Mapping playbook workflows using tools like Draw.io to simplify understanding and streamline response actions.



Strategic Impact

Within the local context this initiative provides significant value to both public and private sector organizations by:

- **Maximizing Defense Capabilities:** Enhancing cybersecurity posture within available budget and infrastructure constraints.
- **Reducing External Reliance:** Minimizing reliance on expensive or overly complex third-party tools that are difficult to tailor.
- **Building a Scalable Foundation:** Serving as an extensible framework adaptable to regional demands and future security initiatives.

Expected Outcomes

- Accelerating incident response times and improving overall handling efficiency.
- Reducing operational overhead by automating repetitive, routine tasks.
- Freeing security analysts to concentrate on proactive threat hunting and strategic analysis.
- Creating better cohesion and interoperability across existing cybersecurity tools.



Literature Review & Background Study

In response to escalating cyber threats and increasingly sophisticated attack vectors, modern enterprises are turning to Endpoint Detection and Response (EDR) alongside Security Orchestration, Automation, and Response (SOAR) solutions to power their security operations. This project aligns with these industry trends, drawing on research and practical methodologies to demonstrate how blending intelligence with automation delivers sustainable cybersecurity.

LimaCharlie's Role in EDR Frameworks

- **Comprehensive Endpoint Telemetry:** LimaCharlie acts as a robust EDR engine capable of harvesting and processing endpoint telemetry in real time to flag anomalous behavior.
- **Granular Threat Detection:** It supports deep process monitoring and detection mechanics—such as identifying credential-harvesting tools like LaZagne—to trigger immediate security alerts.



Tines' Role in SOAR Architecture

- **Dynamic Workflow Automation:** Tines offers a highly adaptable environment for building response playbooks driven by custom logic and dynamic triggers.
- **Low-Code Accessibility:** Its accessible design allows SOC analysts and operational teams to design workflows without requiring complex software engineering skills.
- **Ecosystem Integration:** It seamlessly hooks into communication tools like Slack to dispatch real-time incident notifications and prompt analyst actions.

Global Security Trends & Recommendations

Industry analysis from leading cybersecurity research firms such as Gartner and Palo Alto Networks underscores the impact of pairing EDR with SOAR:

- **Drastic Response Reduction:** Decreases Incident Response (IR) lead times by 60% to 80%.
- **Resource Optimization:** Eliminates repetitive tasks, allowing analysts to focus on high-priority threats.
- **Centralized Visibility:** Aggregates telemetry across distinct tools into a single, cohesive view.



Local and Regional Relevance

Organizations operating face distinct operational hurdles, including constrained cybersecurity budgets and legacy infrastructure. Implementing an automated, cost-effective framework built on adaptable platforms offers clear advantages:

- **Cost Efficiency:** Reduces reliance on expensive proprietary enterprise suites.
- **Talent Empowerment:** Allows local technical teams to gain practical experience with modern SOC tools.
- **Scalable Resilience:** Establishes a centralized response architecture capable of mitigating complex, multi-stage threats.

Analytical Study

This study evaluates the design and implementation of an automated security orchestration framework combining LimaCharlie (EDR) and Tines (SOAR) to build a responsive, self-healing detection mechanism.

Technical Analysis

- **Strengths:** Built on adaptable, cloud-native components that scale effortlessly without requiring fundamental architecture overhauls. Streaming real-time telemetry from LimaCharlie accelerates detection cycles and sharpens analytical precision.
- **Operational Scenarios:** Custom playbooks engineered in Tines handle automated response flows—including host isolation, analyst dispatching, and centralized event logging. Visual workflows designed via Draw.io and integrated into Slack ensure operational transparency.
- **Functional Integration:** API-driven coupling between EDR and SOAR platforms yields a unified, data-informed defense system. Centralizing incident

handling minimizes operational friction and maintains absolute visibility across attack surfaces.

Operational Efficiency & Feasibility

- **Practical Deployment:** The lightweight footprint allows seamless integration into resource-constrained infrastructures, making it highly viable for environments like Ours.
- **Process Automation:** Automating triage and response lightens the burden on SOC teams, delivering faster, consistent outcomes without requiring continuous manual oversight.
- **Data-Driven Strategy:** Aggregating context across endpoints improves decision quality and speeds up containment efforts.



Strategic & Regional Impact

- **Self-Reliance:** Promotes sovereign capability development by reducing reliance on costly commercial tools.
- **Cross-Sector Scalability:** Easily tailors to public, financial, or educational sectors due to its modular design.
- **Capacity Building:** Enables local talent to quickly master and adapt modern SOC technologies.

Challenges and Mitigation Strategies

Challenge	Mitigation Strategy
Skill Gaps	Design intuitive user interfaces and develop thorough operational documentation.
Timeline Constraints	Map out structured workflows and prioritize critical features.
Data Protection	Enforce native security controls, strict RBAC, and credential masking across both tools.
System Interoperability	Standardize API communication protocols and validate data mappings early.

Comprehensive Risk Assessment

Technical Risks

Risk Factor	Analysis	Mitigation Strategy
Platform Compatibility	Data format mismatches or API limits between LimaCharlie and Tines.	Conduct early integration tests and strictly follow vendor API specifications.
Limited Community Support	Community-driven tools may lack instant dedicated support channels.	Maintain internal step-by-step documentation and actively engage tech communities.
Breaking Platform Updates	Unexpected vendor updates altering API endpoints or payload schemes.	Implement configuration version control and monitor upstream release notes.

Operational Risks

Risk Factor	Analysis	Mitigation Strategy
Time Constraints	Single-semester timeframe limits extensive live environment testing.	Adhere to a strict project schedule focused on core functional milestones.
Resource Bottlenecks	Solo execution increases overall workload and single-point responsibility.	Segment implementation into phase-gated deliverables with self-help guides.
Scalability Bottlenecks	Architecture may face ingestion bottlenecks as endpoint counts scale.	Architect playbooks modularly to support horizontal scaling and high throughput.

Security Risks

Risk Factor	Analysis	Mitigation Strategy
Exposure of Sensitive Data	Telemetry and playbooks handle sensitive credentials and host activity logs.	Enforce end-to-end encryption and apply principle-of-least-privilege access.
Configuration Vulnerabilities	Misconfigured response logic could expose attack vectors.	Perform routine security audits and align playbook design with OWASP frameworks.
Platform Downtime	Failure of either tool breaks the automated containment pipeline.	Maintain offline workflow backups and implement graceful failover mechanisms.

Proactively mitigating these technical, operational, and security risks ensures high platform availability while demonstrating sound engineering discipline within modern cybersecurity environments.

System Design & Practical Implementation

The design and implementation phase transitions theoretical concepts into a functional, deployed architecture linking LimaCharlie and Tines. The setup relies on a localized VMware virtual environment to safely simulate real-world attacks, evaluate automated incident response workflows, and eliminate risk to production assets.

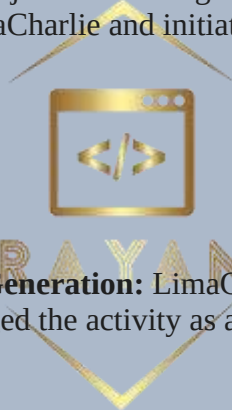
Testbed Setup & Environment Provisioning

A dedicated target virtual machine was deployed within VMware to host threat simulation tools and log endpoint behaviors. Operating inside an isolated virtual lab ensured fine-grained administrative control over security policies while delivering a stable, reproducible testing ground.

Threat Emulation & Execution Mechanics

To trigger real-world security events, the credential-harvesting tool **LaZagne** was executed on the target host. The objective was to generate distinct adversary telemetry that would immediately alert LimaCharlie and initiate downstream response automation.

Upon executing LaZagne:



- **Detection & Telemetry Generation:** LimaCharlie captured suspicious execution events and flagged the activity as a high-severity threat.
- **Webhook Event Forwarding:** The alert payload was transmitted instantly to Tines via secure API webhooks.
- **Orchestrated Playbook Execution:** Tines triggered a multi-stage response sequence:
 - Network isolation of the compromised host within the VMware environment.
 - Real-time dispatch of rich context alerts to the security team via Slack.

- Automated creation and update of incident logs in the central ticketing ledger.

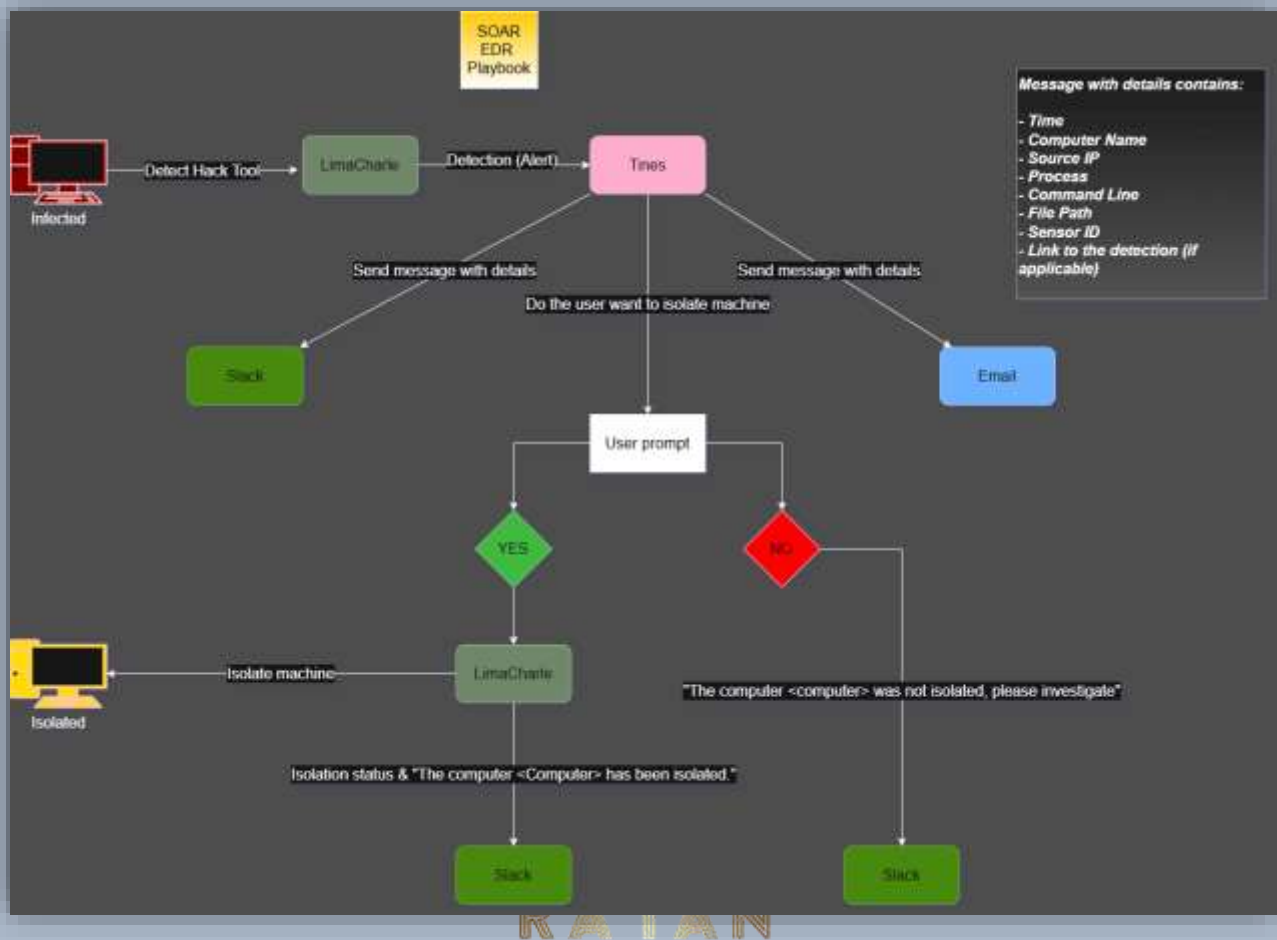
Playbook Workflow Design

A visual playbook workflow was mapped using Draw.io to diagram the end-to-end detection and response pipeline. The blueprint details every state transition from initial execution through automated remediation.

Workflow Components:

- **Node 1 (Target Host):** Compromised guest VM running within VMware.
- **Node 2 (Telemetry Ingestion):** Execution of LaZagne captured and processed by the LimaCharlie sensor.
- **Node 3 (Event Dispatch):** LimaCharlie identifies malicious behavior and sends event payloads to Tines.
- **Node 4 (SOAR Playbook Actions):**
 - Immediate endpoint isolation.
 - Slack notification generation.
 - Incident logging and record-keeping.





Design Phase Outcomes & Achievements

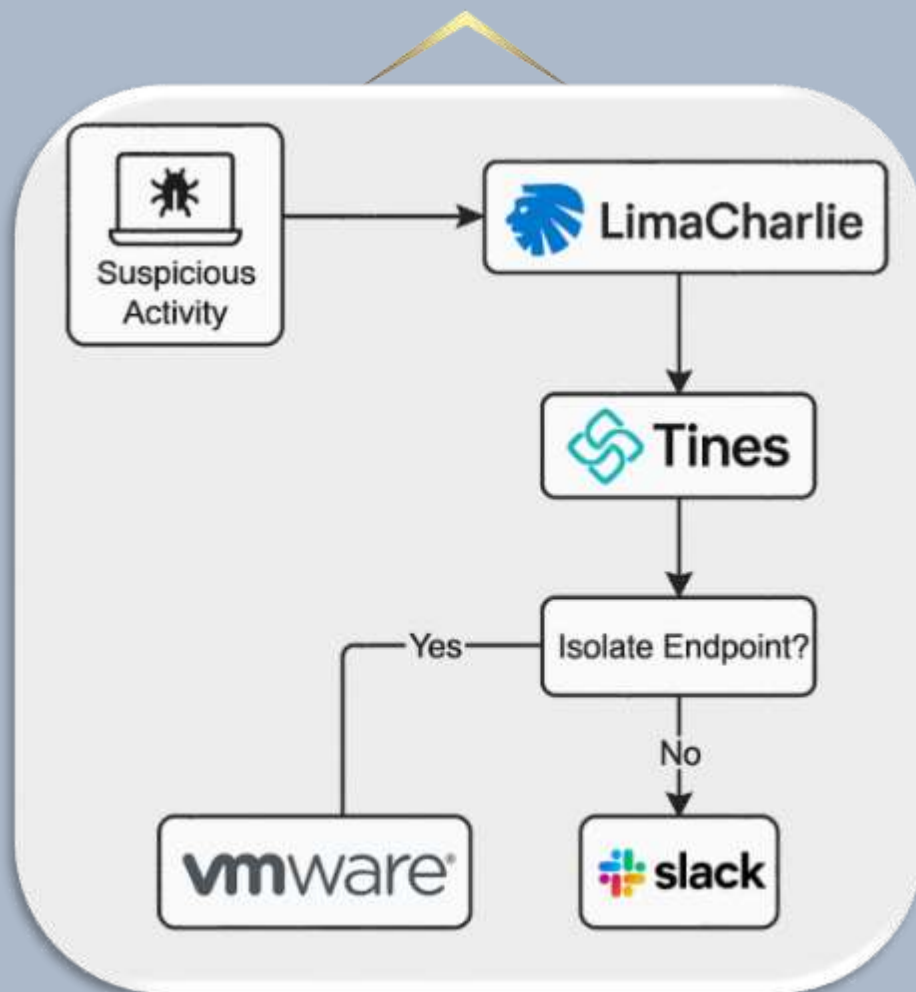
- **Real-World Attack Simulation:** Successfully validated an internal compromise scenario inside a secure, controlled sandbox environment.
- **Automated Threat Processing:** Verified the system's ability to seamlessly capture, analyze, and enrich raw endpoint telemetry upon threat execution.
- **Response Efficiency:** Confirmed the effectiveness of automated response playbooks in drastically minimizing Mean Time to Respond (MTTR).

- **Cross-Platform Interoperability:** Evaluated the end-to-end integration, ensuring consistent data formatting and reliable communication between the EDR, SOAR, and notification channels.

Implementation and Testing

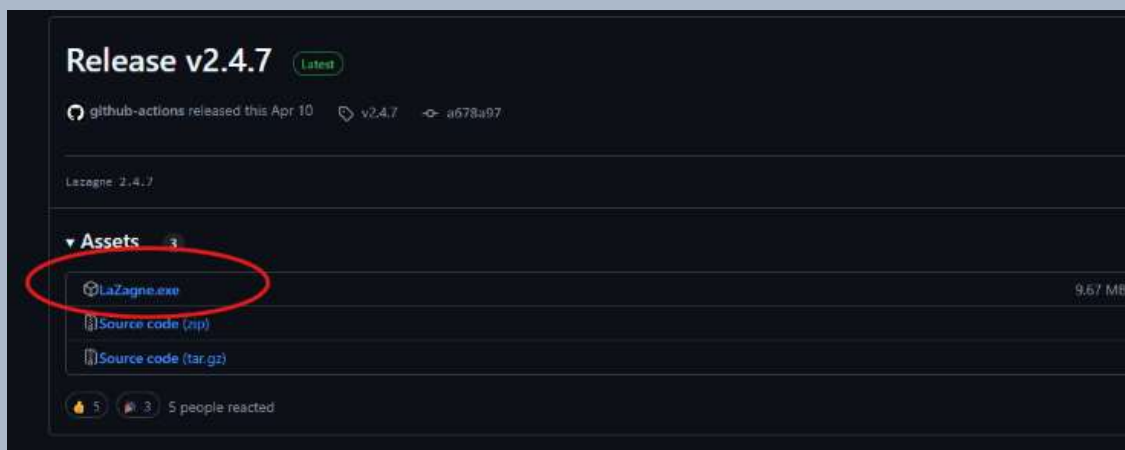
System Execution & Performance Testing

The security framework was deployed within a controlled VMware virtual environment to simulate an end-to-end automated incident response scenario following the execution of exploitation tools. This phase aimed to rigorously test the detection, communication, and response capabilities of LimaCharlie and Tines in executing the designed playbook workflows without manual intervention.



Target System Setup & LaZagne Execution

To simulate host compromise, LaZagne was downloaded directly from its official GitHub repository strictly for testing purposes within the controlled environment:



The LaZagne executable was launched within the target virtual machine to generate anomalous process behavior for LimaCharlie to detect, harvest, and analyze.

A screenshot of a Windows PowerShell terminal window titled "Administration: Windows PowerShell". The terminal shows the command "PS C:\Users\Administrator\Downloads> .\LaZagne.exe all" being executed. The output of the command is displayed in a blue background with white and green text. It shows the "LaZagne Project" banner, followed by two lines of system masterkey decryption. Then, it shows the "User: SYSTEM" and "Hashdump passwords" section, which lists several system accounts and their hashes. Finally, it shows the "Lsa_secrets passwords" section, which lists several system secrets and their hashes. The terminal ends with the message "[!] 0 passwords have been found." and the elapsed time "1.6875920295715332".

```
Administration: Windows PowerShell
For more information launch it again with the -v option
elapsed time = 1.7499415874481201
PS C:\Users\Administrator\Downloads> .\LaZagne.exe all

The LaZagne Project
| BANG BANG |

[+] System masterkey decrypted for 3e8365e3-19cf-4451-95f2-87ce8ccb3949
[+] System masterkey decrypted for 855e2624-87ac-40cf-8c9f-61b7fa0313e5

##### User: SYSTEM #####

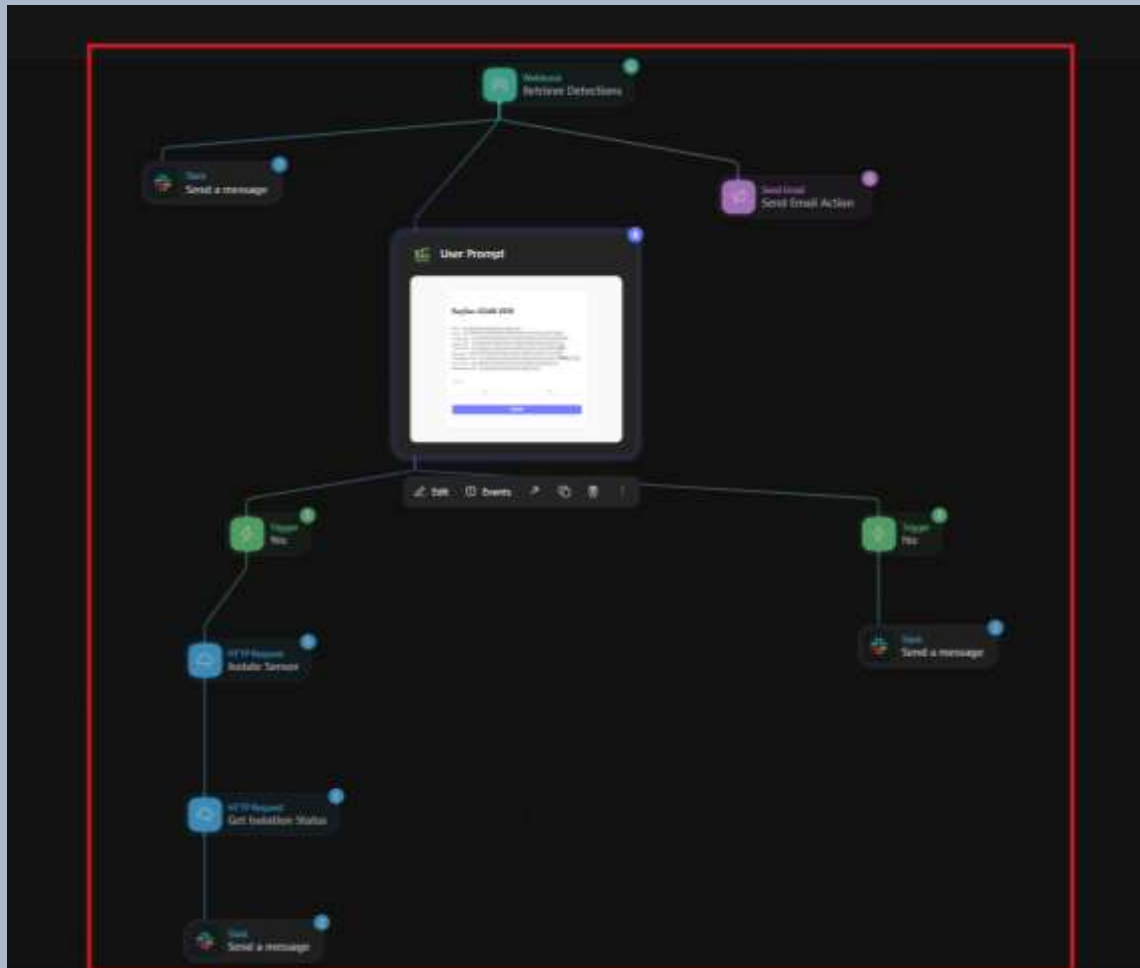
----- Hashdump passwords -----
Administrator:500:aad3b435b51404eeaad3b435b51404ee:0bdee0b1717f01059e8648a1c682ec22:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:7c6a4f04f89aaf6a5101da1bde84cf50:::

----- Lsa_secrets passwords -----
DPAPI_SYSTEM
0000 01 00 00 00 BB 5E 1C 27 45 E6 8E 75 36 36 05 DD .....^.'E..u66..
0010 80 14 83 3E A6 9D 5F 56 86 D2 E1 A0 17 85 23 86 ...>..._V.....#..
0020 AD 19 AA 7E 2A A8 BF 99 3F 01 3C B6 ...~*...?.<..

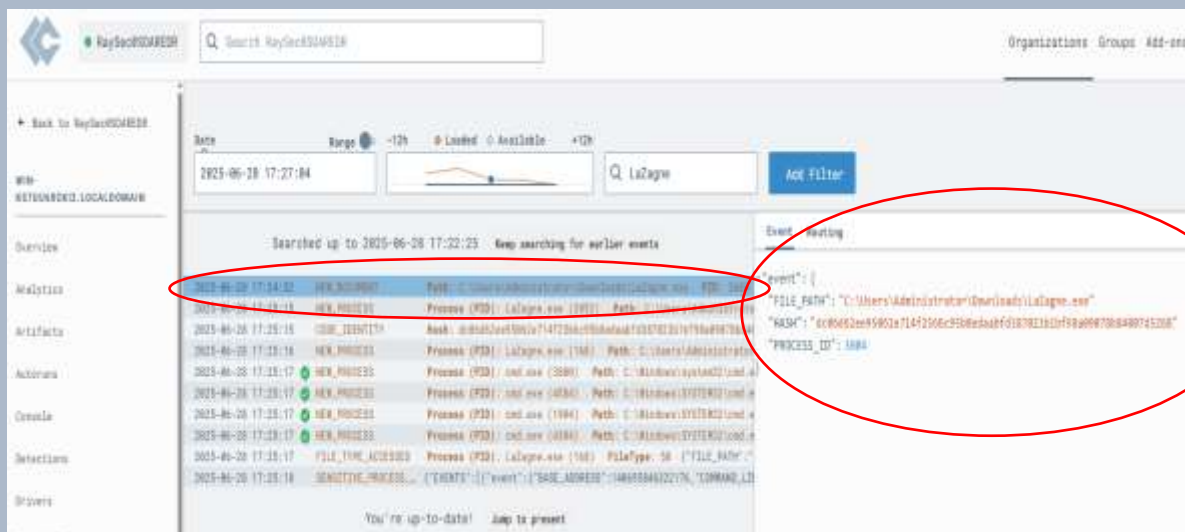
NL$KM
0000 40 00 00 00 00 00 00 00 00 00 00 00 00 00 00 @.....
0010 C3 F8 3D C5 A5 56 90 88 45 B3 81 85 DC 48 06 ED ..=..V..E...H..
0020 DF D6 00 94 8E A9 13 BD 42 1E FD D6 47 E5 74 7D ...:..B...G.t}
0030 05 7D 70 0F 40 66 F7 68 22 87 6D 53 13 34 CF 80 ...]p.@f.k".mS.4..
0040 67 E8 68 8A 1A 38 CA AB DF AA 06 8B 17 46 A8 04 g.k..8.....F..
0050 8D 81 9F A2 18 19 48 76 99 6A 5B 37 89 20 4C 07 .....Hv.}[7; L.

[!] 0 passwords have been found.
For more information launch it again with the -v option
elapsed time = 1.6875920295715332
PS C:\Users\Administrator\Downloads>
```

Upon executing the tool, real-time telemetry was immediately streamed to LimaCharlie, which flagged the activity as an active threat. Detecting this event automatically triggered the API integration with Tines to initiate the response workflow.



As observed below, the event was successfully captured by LimaCharlie:



Slack & Email Alerting

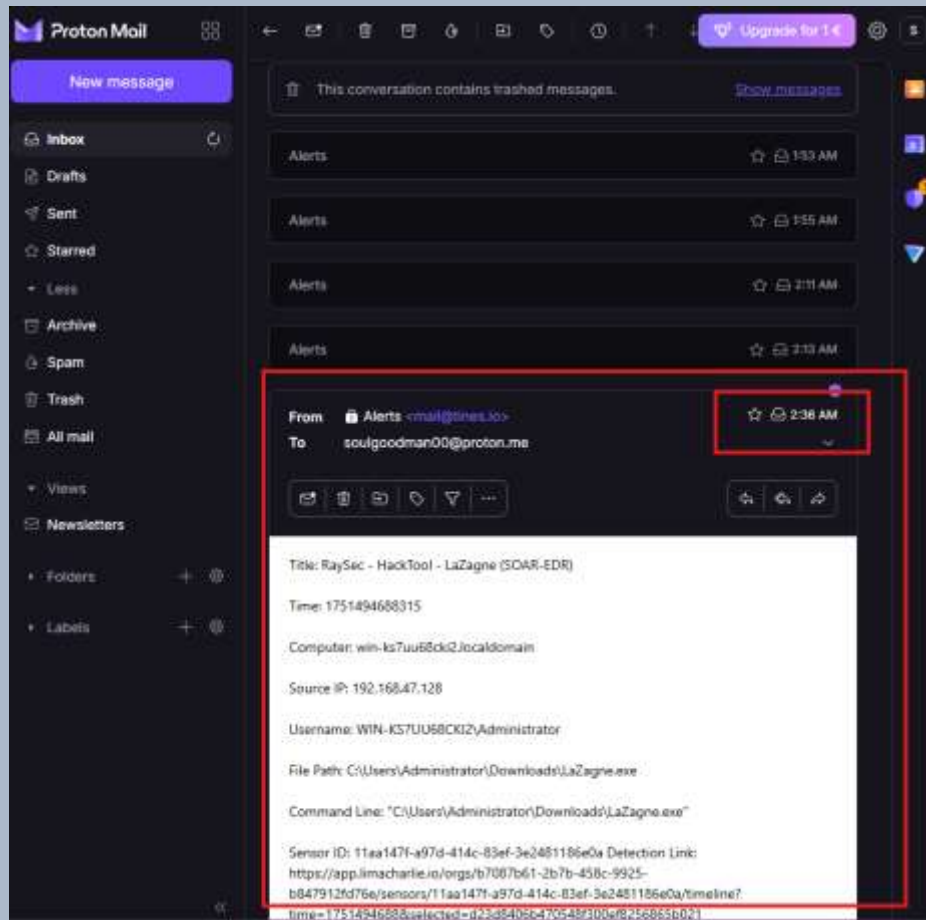
Once the event payload reaches Tines, the system automatically dispatches a real-time notification to a designated Slack channel, alongside an email alert sent to a pre-configured administrator account.



Slack Notification Output:



Email Alert Output:



The notifications contain critical threat context—including the tool name, host details, timestamp, and full execution parameters as detailed below—empowering the security team to make informed containment decisions:

Both Slack and email alerts include:

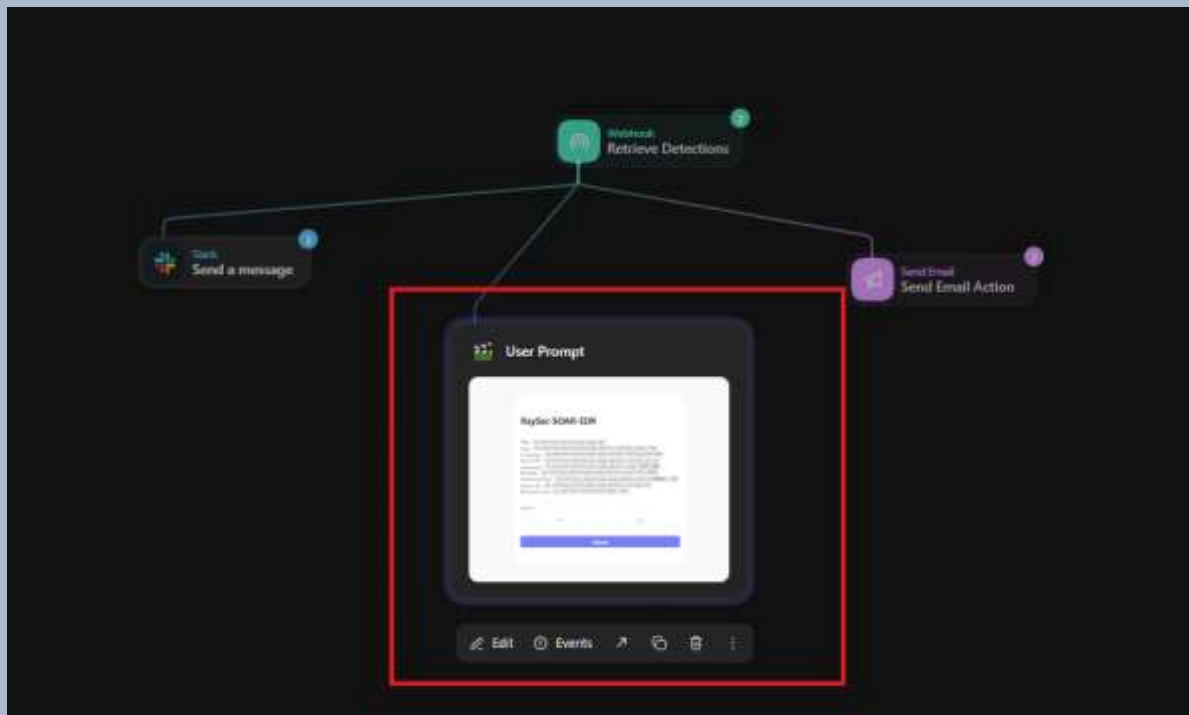
- Title
- Timestamp
- Hostname / Computer Name

- Source IP
- Process Name
- Executed Command Line
- File Path
- Sensor ID
- Direct link to the detection entry (where applicable)



Interactive User Prompt in Tines

Following the initial alert dispatch, Tines triggers an interactive User Prompt interface for the security analyst, presenting an immediate choice to either isolate the host or dismiss the event.



If the analyst selects **"Isolate Host"**, Tines immediately executes containment procedures by issuing direct network isolation commands to LimaCharlie.

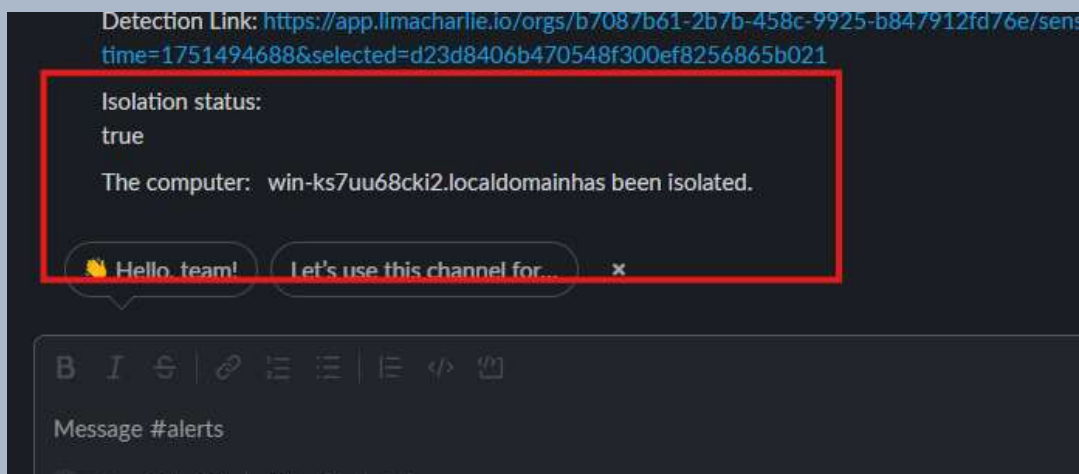
RaySec-SOAR-EDR

Title: RaySec - HackTool - LaZagne (SOAR-EDR)
Time: 175149468315
Computer: win-4s7uu68ck2.localdomain
Source IP: 192.168.47.128
Username: WIN-KS71AUG8CKQ2\Administrator
File Path: C:\Users\Administrator\Downloads\LaZagne.exe
Command Line: "C:\Users\Administrator\Downloads\LaZagne.exe"
Sensor ID: 11aa147f-a97d-414c-83ef-3e2481186e0a
Detection Link: <https://app.limacharlie.io/>

Isolate?

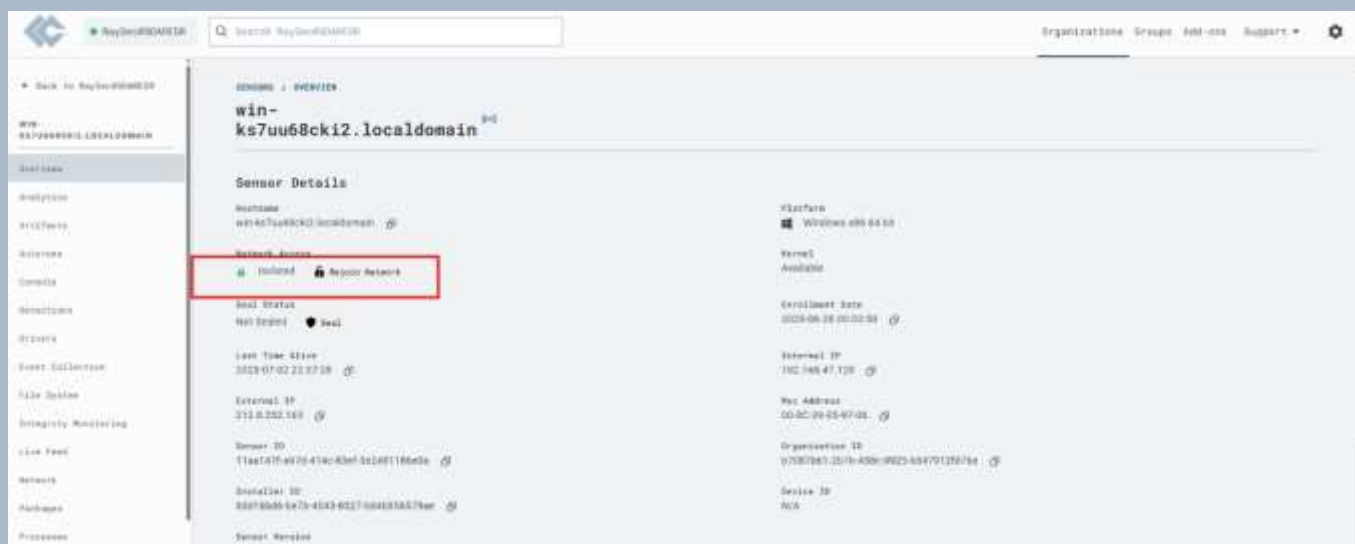
Isolation Confirmation via LimaCharlie, VMware, and Slack

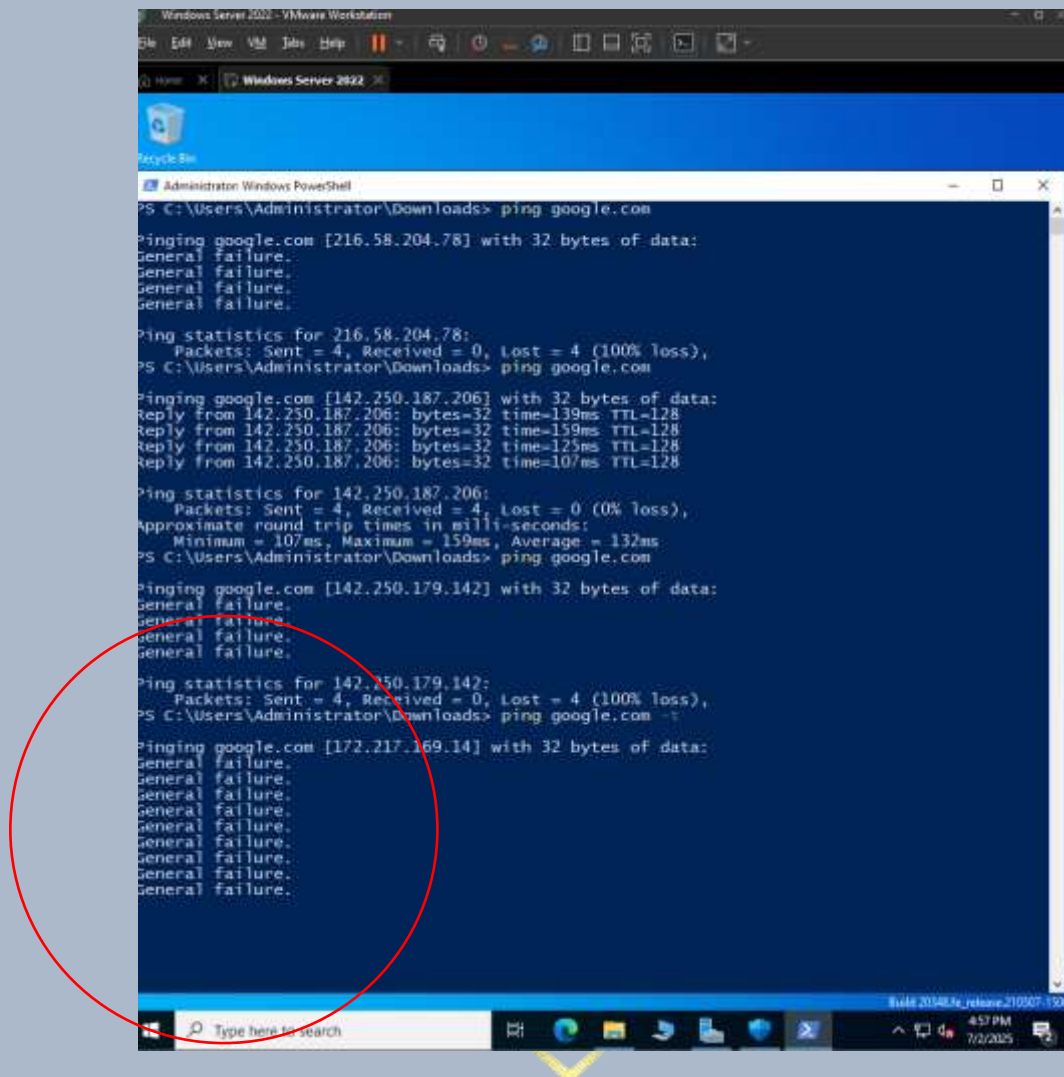
Slack Confirmation Output:



Once isolation is enforced, the host status in LimaCharlie updates to **"Isolated"**, instantly cutting off network and internet access for the virtual machine.

Subsequent attempts to execute network commands or establish internet connectivity failed on the target host, confirming that endpoint isolation was successfully enforced:





This confirms the success of the security procedure and the effectiveness of the automated response pipeline.

Host Isolation Dismissal & Alternative Alerting

If the analyst decides against isolating the system, Tines sends a follow-up notification to Slack confirming that the host was left online in accordance with the analyst's selection.

This confirms the success of the security procedure and the effectiveness of the automated response pipeline.

RaySec-SOAR-EDR

Title: RaySec - HackTool - LaZagne (SOAR-EDR)
Time: 1751494688315
Computer: win-ks7uu68ck02.localdomain
Source IP: 192.168.47.128
Username: WIN-KS7UU68CK02\Administrator
File Path: C:\Users\Administrator\Downloads\LaZagne.exe
Command Line: "C:\Users\Administrator\Downloads\LaZagne.exe"
Sensor ID: 11aa147f-a97d-414c-83ef-3e2481186e0a
Detection Link: <https://app.lmachahe.io/r...21>

Isolate?

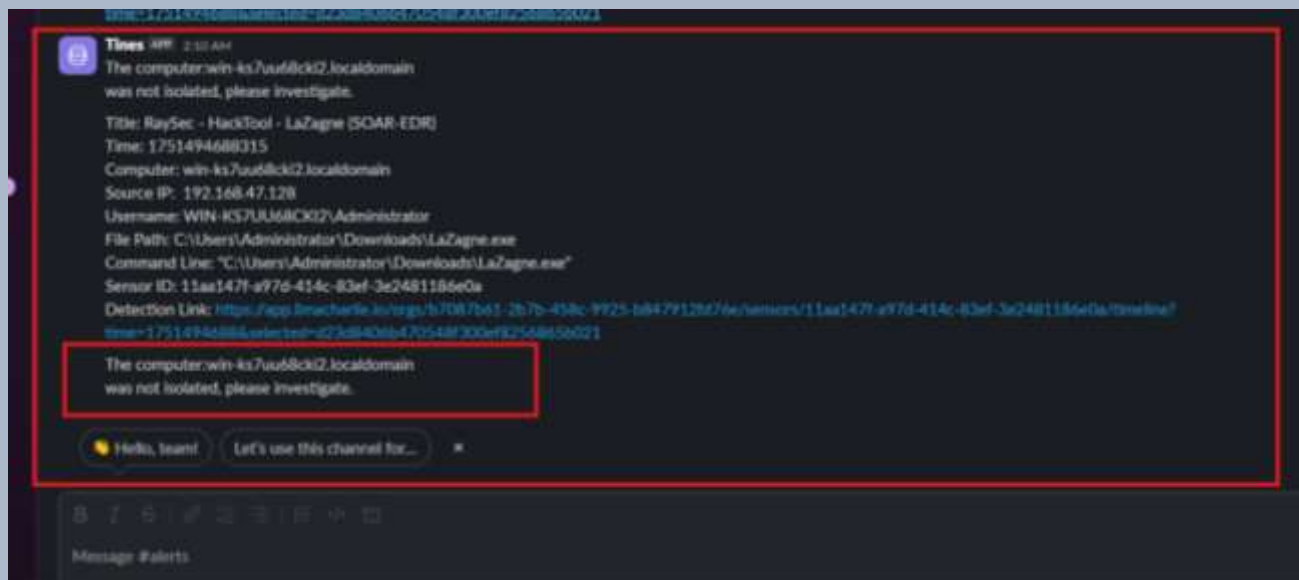
Yes

No

Submit



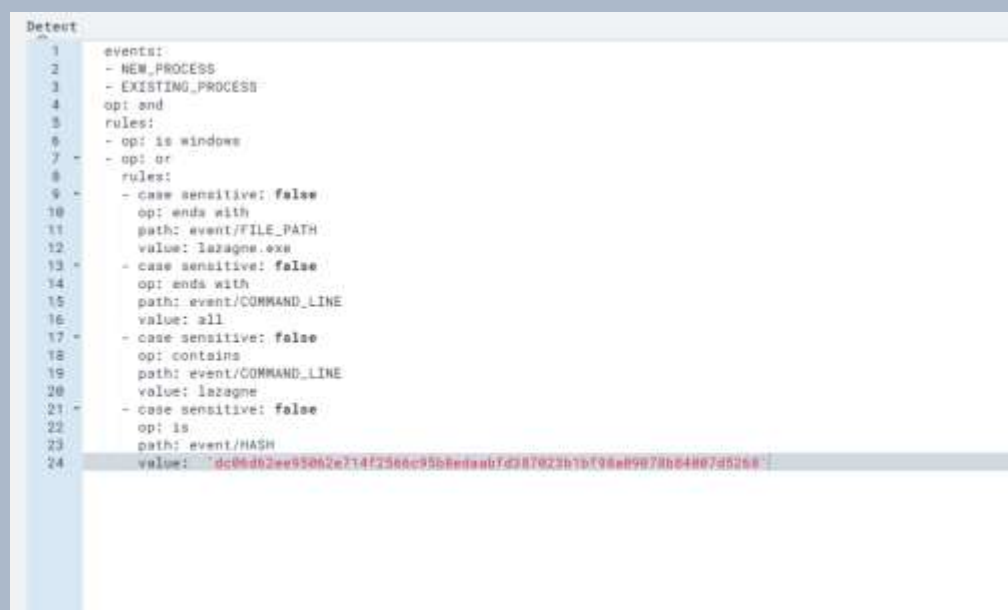
Slack Dismissal Output:



Detection and Response (D&R) Rules in LimaCharlie

Detection and Response rules were explicitly programmed within LimaCharlie using structured YAML syntax to detect credential-dumping utilities (such as LaZagne) and immediately trigger notification playbooks upon execution.

Detection Rules (Detect Block):



Detection Logic Breakdown

Response Rules (Respond Block):

```
Response
1 - action: report
2 metadata:
3   author: RaySec@SOAEDR
4   description: LaZagne (SOAR-EDR-Tool)
5     from view
6   falsepositives:
7     - Unlikely alerts
8   level: medium
9   tags:
10    - attack.credential_access
11    name: RaySec - HackTool - LaZagne (SOAR-EDR)
```

Response Rules Logic Breakdown

When the conditions defined in the detection block are met, LimaCharlie executes the following response action:



- **Action Type:** Generates a structured detection report entry. It intentionally avoids immediate automated endpoint isolation at the EDR layer, passing execution control to Tines for orchestrated human-in-the-loop triage.

- **Metadata & Context:**

- **Author:** RaySec@SOAEDR
- **Description:** Alert triggering on LaZagne credential-recovery execution (SOAR-EDR project).
- **False Positives:** Low likelihood / Not expected in normal baseline operations.
- **Severity Level:** Medium

- **Tags & Framework Alignment:** Categorized under the **MITRE ATT&CK** framework as Credential Access (`attack.credential_access`).
- **Detection Name:** RaySec - HackTool - LaZagne (SOAR-EDR)

Implementation & Testing Validation Notes

- **Dual-Path Verification:** Both the isolation and non-isolation workflows were fully validated to ensure fault-tolerant decision processing within the Tines orchestration engine.
- **Performance Metrics:** Metrics were gathered on alert latency, context fidelity, and execution speed across the virtualized testbed to ensure low-latency response times.
- **Visual Documentation:** Every stage of execution—from threat injection to host network isolation—was visually captured and recorded as part of the project's practical evidence documentation.



Conclusion and Future Horizons

Project Conclusion

This project represents a practical step toward engineering an automated cybersecurity framework utilizing modern paradigms in incident orchestration, context-rich analysis, and response. By integrating LimaCharlie (EDR) with Tines (SOAR), the system establishes a functional model capable of autonomously detecting suspicious endpoint behavior and orchestrating response workflows. This architecture directly improves Security Operations Center (SOC) efficiency, reduces Mean Time to Respond (MTTR), and minimizes operational friction for security analysts.

Controlled validation within a VMware-virtualized sandbox confirmed the operational viability of the pipeline. Upon executing the LaZagne credential-recovery utility, the system successfully triggered the end-to-end detection sequence—ranging from initial telemetry ingestion to conditional analyst prompts and eventual endpoint isolation. Supporting integrations with Slack and structural flow mapping via Draw.io ensured operational clarity and streamlined incident communication.

Future Horizons & Strategic Enhancements

The current implementation establishes a resilient foundation for future scaling across several key domains:

- **Extended Platform Integration:** Integrating central Security Information and Event Management (SIEM) solutions and external Threat Intelligence Platforms (TIPs) to enrich alerts with deeper contextual telemetry.
- **Predictive & Behavioral Analytics:** Incorporating machine learning techniques to identify anomalous behavior patterns and project threat trajectories prior to exploitation.
- **Expanded Playbook Ecosystem:** Developing targeted orchestration playbooks to address complex threat vectors, including spear-phishing campaigns, active directory compromise, and wireless network intrusion.

- **Performance Metrics & KPI Tracking:** Standardizing real-time dashboard telemetry to monitor Key Performance Indicators (KPIs) such as triage accuracy, MTTR, and false-positive rates.
- **Analyst User Experience Optimization:** Refining interactive triage interfaces within Tines to streamline decision-making for junior analysts and reduce cognitive overload during high-volume alert events.

User Guide: LimaCharlie & Tines Automated Incident Response System

This guide outlines the operational procedures for managing the automated security detection and incident response framework. It is designed to assist security analysts in executing test scenarios, monitoring live alerts, and making containment decisions.

System Prerequisites

- **LimaCharlie EDR:** Active account with the LimaCharlie sensor deployed on one or more monitored endpoints.
- **Tines SOAR:** Active account configured with the custom RaySec incident response playbook.
- **Alert Channels:** Integrated Slack workspace channel and administrative email inbox linked to Tines webhooks.
- **Sandbox Environment:** VMware workstation/hypervisor instance hosting the target endpoint for threat injection and isolation validation.

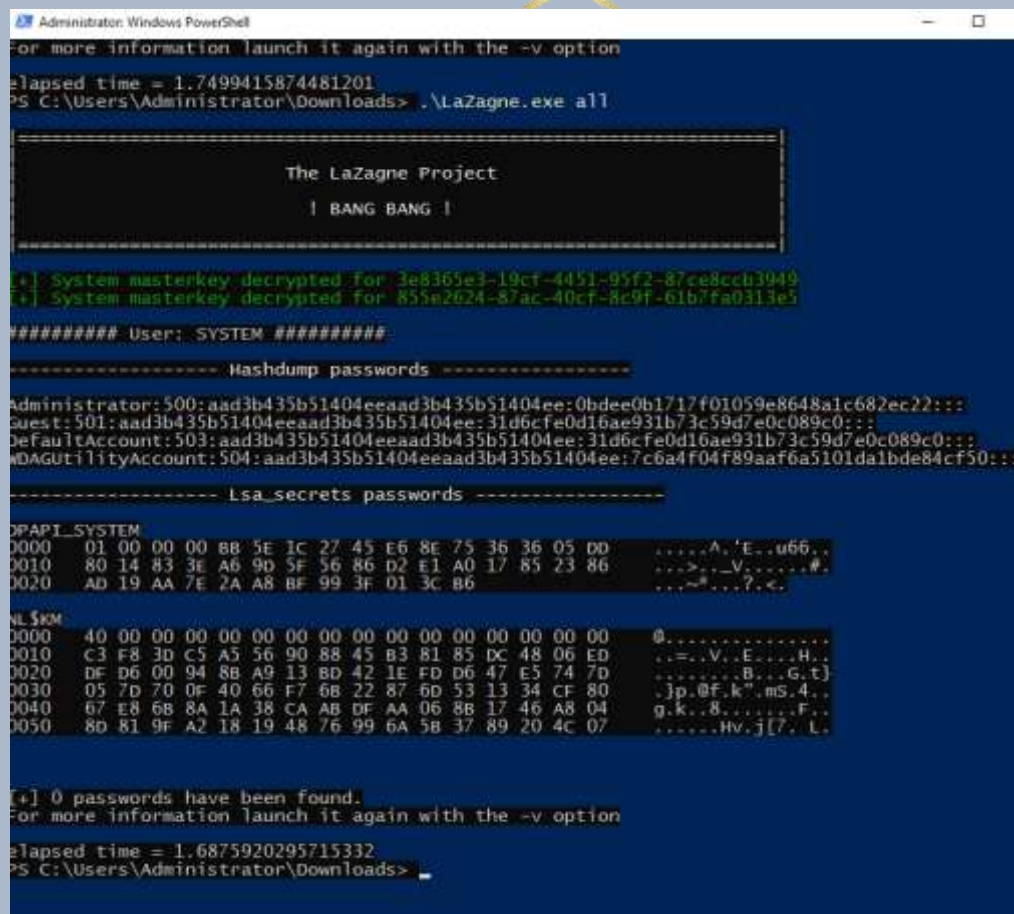


Operational Workflow

Step 1: Endpoint Monitoring & Threat Simulation

1. Launch the target virtual machine within the VMware environment.
2. Execute the credential-dumping utility (LaZagne.exe) within the target guest environment to trigger telemetry collection.
3. The host-installed LimaCharlie sensor captures process execution telemetry in real time and evaluates it against the active Detection & Response (D&R) rules.

LaZagne Execution in VMware:



```
Administrator: Windows PowerShell
for more information launch it again with the -v option
elapsed time = 1.7499415874481201
PS C:\Users\Administrator\Downloads> .\LaZagne.exe all

=====
The LaZagne Project
| BANG BANG |
=====

[+] System masterkey decrypted for 3e8365e3-19cf-4451-95f2-87ce8ccb3949
[+] System masterkey decrypted for 855a2624-87ac-40cf-8c9f-61b7fa0313e5

##### User: SYSTEM #####

----- Hashdump passwords -----
Administrator:500:aad3b435b51404eeaad3b435b51404ee:0bdee0b1717f01059e8648a1c682ec22:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:7c6a4f04f89aaf6a5101da1bde84cf50:::

----- Lsa_secrets passwords -----
DPAPI_SYSTEM
0000 01 00 00 00 8B 5E 1C 27 45 E6 8E 75 36 36 05 DD .....A..E...u66..
0010 80 14 83 3E A6 9D 5F 56 86 D2 E1 A0 17 85 23 86 ...>...V.....#..
0020 AD 19 AA 7E 2A A8 BF 99 3F 01 3C B6 .....~...?.<..

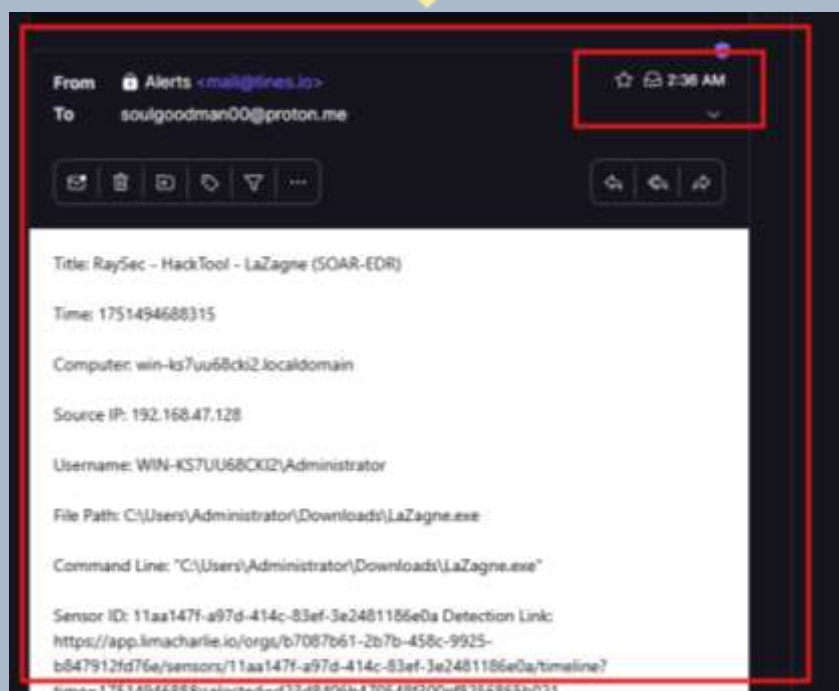
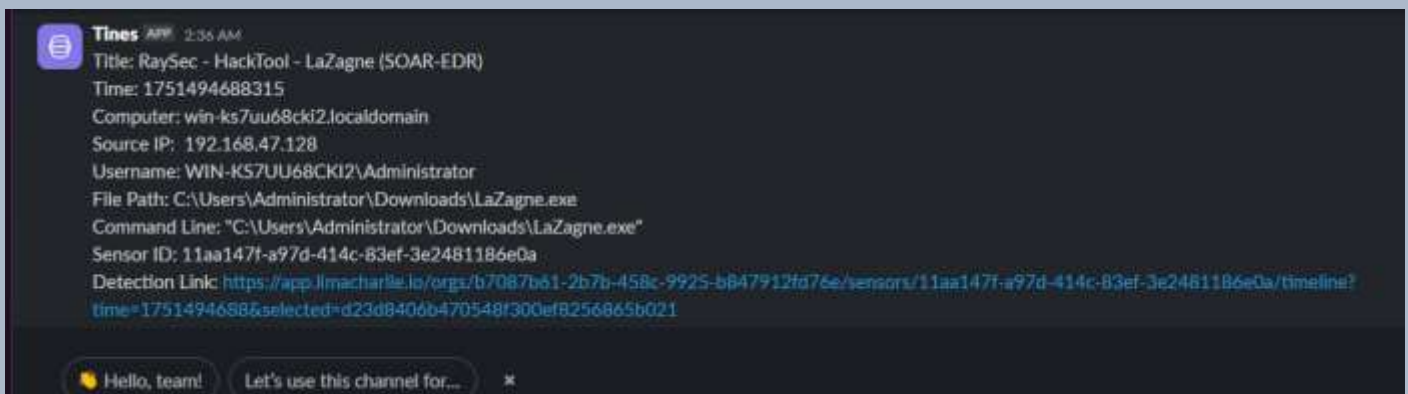
NL$KM
0000 40 00 00 00 00 00 00 00 00 00 00 00 00 00 00 @.....
0010 C3 F8 3D C5 A5 56 90 88 45 B3 81 85 DC 48 06 ED ...=.V..E...H..
0020 DF D6 00 94 8B A9 13 BD 42 1E FD D6 47 E5 74 7D .....B...G.t}
0030 05 7D 70 0F 40 66 F7 6B 22 87 6D 53 13 34 CF 80 ..}p.@f.k".ms.4..
0040 67 E8 6B 8A 1A 38 CA AB DF AA 06 8B 17 46 A8 04 g.k..8.....F..
0050 8D 81 9F A2 18 19 48 76 99 6A 5B 37 89 20 4C 07 .....Hv.j[7..L..

[+] 0 passwords have been found.
for more information launch it again with the -v option
elapsed time = 1.6875920295715332
PS C:\Users\Administrator\Downloads>
```

Step 2: Receiving Alerts via Slack & Email

- Upon detecting the malicious process execution, the framework dispatches an automated notification payload to the designated Slack channel and administrative email inbox via Tines webhooks.
- The alert payload provides critical incident context, including the identified threat name, target host details, timestamp, and process parameters required for triage.

Slack and Email Alert Delivery:



Step 3: Interactive Response Prompt in Tines

- Tines automatically generates an interactive User Prompt interface for the security analyst, requesting an immediate containment decision: **Isolate Host** or **Dismiss**.
- **Option A ("Isolate"):** Selecting this option immediately issues an automated isolation API call to LimaCharlie, cutting off network access to the target endpoint.
- **Option B ("Dismiss"):** Selecting this option cancels the isolation procedure and dispatches a notification confirming that no containment actions were taken per analyst decision.

Interactive User Prompt Interface in Tines:



RaySec-SOAR-EDR

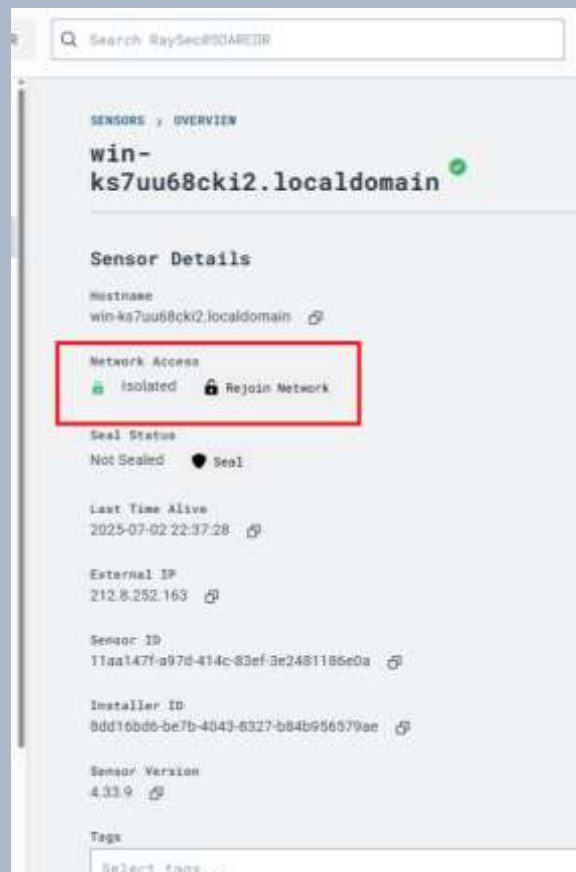
Title: RaySec - HackTool - LaZagne (SOAR-EDR)
Time: 1751494688315
Computer: win-ks7uu68cki2.localdomain
Source IP: 192.168.47.128
Username: WIN-KS7UU68CKI2\Administrator
File Path: C:\Users\Administrator\Downloads\LaZagne.exe
Command Line: "C:\Users\Administrator\Downloads\LaZagne.exe"
Sensor ID: 11aa147f-a97d-414c-83ef-3e2481186e0a
Detection Link: <https://app.limacharlie.io/...21>

Isolate?

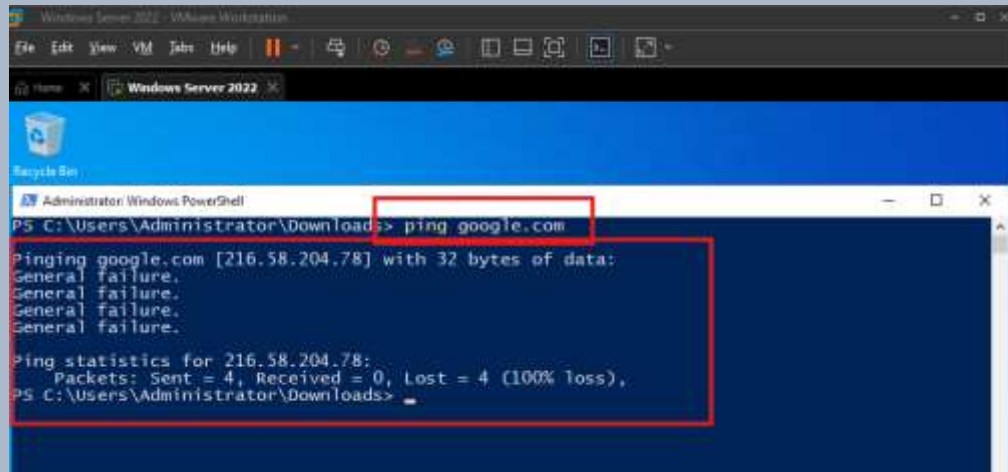
Step 4: Verifying Containment Execution

- **EDR Portal Verification:** If isolation was selected, the endpoint's status in the LimaCharlie dashboard immediately updates to **"Isolated"**.
- **Host-Level Verification:** On the VMware target guest, all outbound and inbound network traffic is blocked, cutting off internet connectivity and isolating the endpoint from the internal network while keeping the LimaCharlie agent control channel active.

LimaCharlie Host Status — "Isolated":



VMware Target Host Post-Isolation — No Internet Access:



Step 5: Post-Dismissal Notification (Non-Isolation Path)

- If the security analyst opts to decline isolation during the interactive prompt, Tines bypasses the network isolation action and routes through the alternative decision branch.
- A follow-up confirmation message is automatically dispatched to the designated Slack channel, logging that the endpoint remained online per manual analyst override.

Slack Notification — Host Isolation Dismissed:

1. Threat Execution: VMware Sandbox.

Execute `LaZagne.exe` on the target Windows guest VM to generate malicious process telemetry.

2. Telemetry Ingestion: LimaCharlie EDR.

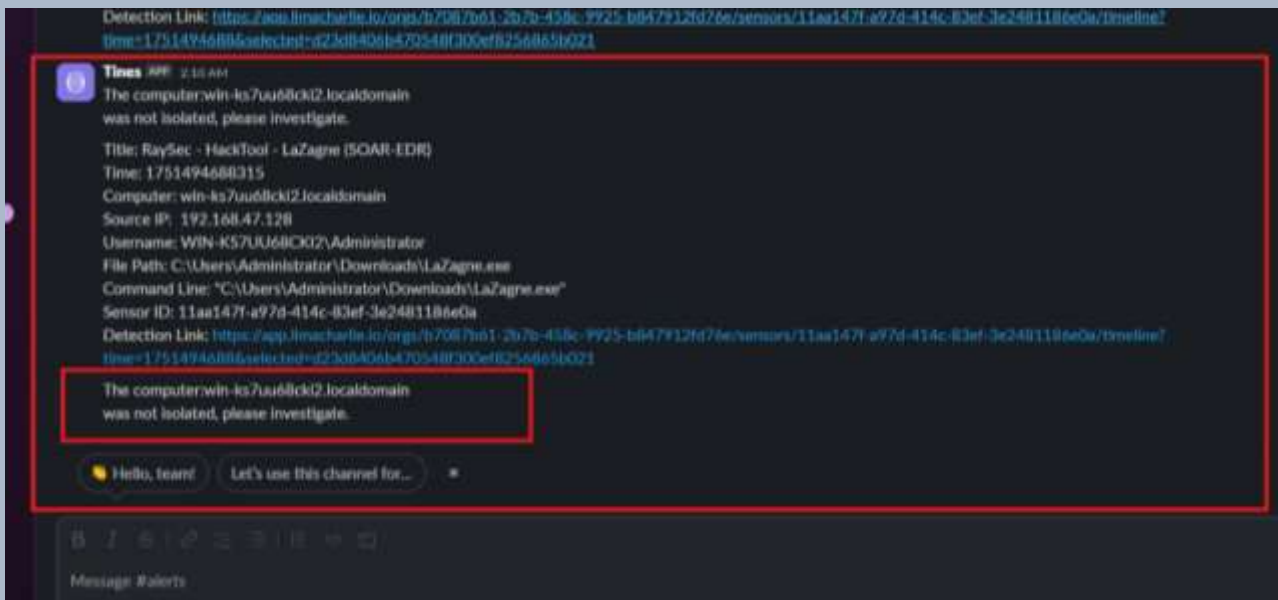
The custom D&R rule detects process indicators (T1003) and sends event metadata to Tines via webhook.

3. Interactive Triage: Tines SOAR / Slack.

Review the alert context dispatched to Slack/Email and make a containment decision via the interactive Tines prompt.

4. Automated Enforcement: Containment / Dismissal.

Choosing **Isolate** enforces immediate network blockade via LimaCharlie; choosing **Dismiss** logs the action to Slack while keeping the endpoint online.



Step 6: Incident Management, Audit Trails & Operational Best Practices

Incident Management & Review in Tines

- **Audit Logs & Execution History:** Security analysts can access the Tines dashboard at any time to review complete execution logs, payload histories, and decision timestamps for every triggered incident.
- **Playbook Extensibility:** The orchestration workflow is fully modular. Security engineers can update the Tines playbook to introduce additional conditional

branching, supplementary notification channels (e.g., Microsoft Teams, PagerDuty), or automated SIEM log indexing as organizational requirements evolve.

Critical Operational Best Practices

- **Integration Health Checks:** Ensure active API tokens and webhook integrations across LimaCharlie, Tines, and Slack are routinely validated to maintain uninterrupted communication pipelines.
- **Response Verification & Performance Monitoring:** Regularly monitor telemetry accuracy, detection latency, and isolation effectiveness within LimaCharlie reporting panels and dedicated VMware sandbox environments.
- **Rapid Analyst Triage:** Interactive security prompts require prompt analyst response to minimize the window of exposure during credential-dumping or active post-exploitation events.



1. *Audit & Management:* Tines Console.

Access Tines execution logs to audit historical incident timelines, payloads, and response choices.

2. *Modular Playbook Scaling:* SOAR Customization.

Extend D&R capabilities by attaching new integration nodes or additional threat intelligence lookup modules.

3. *Continuous Maintenance:* Operations & SLA.

Maintain active API integrations and maintain strict SLA response times to interactive triage prompts.

References & Technical Documentation

The architecture, detection logic, and orchestration workflows in this project are based on the following official documentation and industry standards:

1. LimaCharlie EDR Platform

- Docs & Rule Engine: [LimaCharlie Documentation — Detection & Response \(D&R\) Rules](#)
 - *Application:* Writing conditional logic for system telemetry (`NEW_PROCESS`, `EXISTING_PROCESS`) and configuring direct isolation actions.
- REST API & Telemetry Reference: [LimaCharlie API Documentation](#)
 - *Application:* Integrating host sensors with external webhook listeners (Tines).



2. Tines SOAR Platform

- Automation & Workflow Library: [Tines Platform Documentation](#)
 - *Application:* Designing custom playbooks, handling inbound HTTP webhooks, and building interactive human-in-the-loop prompts.
- Security Orchestration Standards: [Tines Automation Playbooks & Integration Patterns](#)

- *Application:* Automating multi-channel alerts across Slack and email endpoints.

3. MITRE ATT&CK® Framework

- Technique T1003 (OS Credential Dumping): MITRE ATT&CK — Technique T1003
 - *Application:* Mapping LaZagne credential extraction activity under the Credential Access tactic (`attack.credential_access`).
- Tactic TA0006 (Credential Access): MITRE ATT&CK — Credential Access Tactic

4. Security Tools & Testing Environments

- LaZagne Security Tool: Alessandro Zanni — LaZagne Project Repository
 - *Application:* Simulating post-exploitation credential dumping to test rule coverage and pipeline latency.
- VMware Virtualization: VMware Workstation Pro Documentation
 - *Application:* Hosting the isolated sandbox environment to safely test and verify network isolation.